MIL-E-1/1571B
26 March 1973

SUPERSEDING
MIL-E-1/1571A
30 November 1965

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE 8467

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Reflex, tunable frequency range 12.4 to 14.5 GHz, waveguide output, integral cavity.

ABSOLUTE RATINGS:

Parameter:	Ef	Ers	Er	Tuner Plate Temperature
Unit:	V	Vdc	Vdc	°C
Maximum:	7.0	625	-1,000	160
Minimum:	5.7	---	-20 (See Note 1)	---

PHYSICAL CHARACTERISTICS:

Dimensions:	See Figure 1
Base:	A3-108
Cathode:	Unipotential

TEST CONDITIONS:

Parameter:	Ef	Ers	Er	F	Tuner Plate Temperature
Unit:	V	Vdc	Vdc	GHz	°C
Test 1:	6.3	600	-140 to -400	12.4 & 14.5	100 (See Note 2)
Test 2:	6.3	300	-85 to -220	12.4 & 14.5	100 (See Note 2)

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	CONDITIONS	TEST	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1						
-----	Emission	Ef = 5.7 V See notes 3, 4 and 5	1	Δ Irs/ Irs	---	15	%
1301	Heater current	Ef = 6.3 V See note 3		If	---	0.5	A
1256	Resonator current (1)	Er/max Po	1	Irs	---	66	mAdc
1256	Resonator current (2)	Er/max Po	2	Irs	---	35	mAdc
4213	Reflector voltage (1)	Er/max Po	1	Er	-140	-400	Vdc
4213	Reflector voltage (2)	Er/max Po	2	Er	-85	-220	Vdc
4229	Total reflector current	See notes 3, 4 and 5		Ir	---	5	μAdc
4223	Mechanical tuning range	Er/max Po		F	12.4	14.5	GHz
4250	Output power (1)	Er/max Po	1	Po	140	---	mW
4250	Output power (2)	Er/max Po	2	Po	20	---	mW
4280	Electronic tuning range (1)	Er/max Po See note 7	1	Δ F	50	---	MHz
4280	Electronic tuning range (2)	Er/max Po See note 7	2	Δ F	20	---	MHz
	Quality conformance inspection, part 3						
4551	Life-test provisions	Group D Er/max Po	1	t	500	---	hrs
	Life-test end point:						
4250	Output power (1)	Er/max Po	1	Po	112	---	mW

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NOTES:

1. The reflector voltage shall always be negative with respect to the cathode by at least 20 volts.

2. All oscillation tests shall be made with the tube rigidly connected to appropriate waveguide equipment (e.g., RG-107/U, or equivalent). The load VSWR for the tube shall be less than 1.1:1. Forced-air cooling (3.7 lb./min) is required for beam input powers above 10 watts. For maximum tube life, the tube temperature should be less than 100°C. Tube tuner plate temperature shall be measured at the point indicated on figure 1.

3. These tests to be performed immediately after the holding period.

4. The tube shall not be oscillating during the test.

5. The resonator current shall not drop more than the specified percentage when the heater voltage is decreased from the normal operating potential to the lower specified value.

6. After 2 minutes with all voltages applied, total reflector current shall not exceed the specified limits.

7. The electronic tuning range is measured between half-power points.

Custodians:
Army - EL
Navy - EC
Air Force - 80

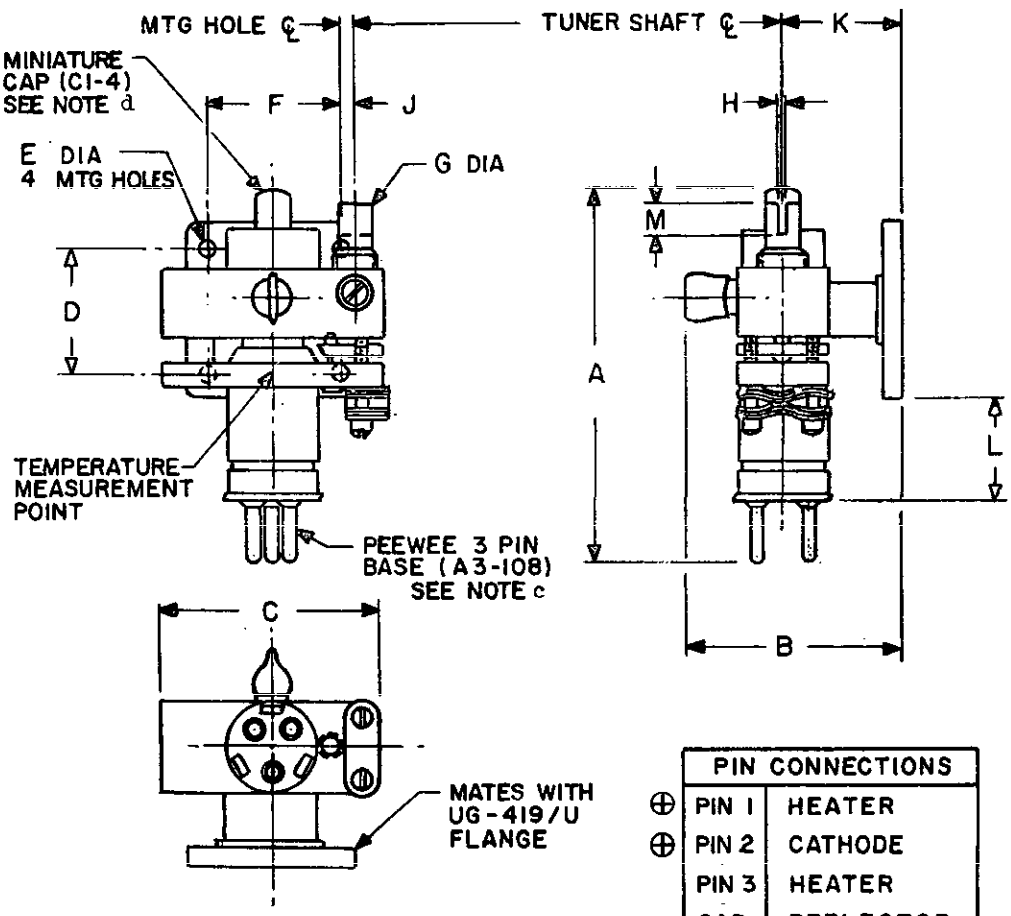
Review activities:
Army - EL, MU
Navy - SH
Air Force - 11,

User activities:
Army - MU
Navy - MC, CG, OS

Preparing activity:
Air Force - 80

(Project 5960-2650)

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PIN CONNECTIONS	
⊕ PIN 1	HEATER
⊕ PIN 2	CATHODE
PIN 3	HEATER
CAP	REFLECTOR

⊕ INTERNALLY CONNECTED

DIMENSIONS												
DIM	INCHES			MILLIMETERS			DIM	INCHES			MILLIMETERS		
	MIN	NOM	MAX	MIN	NOM	MAX		MIN	NOM	MAX	MIN	NOM	MAX
QUALIFICATION INSPECTION							B			1.750			44.45
D	.952	.956	.960	24.18	24.28	24.38	C			1.750			44.45
E	.144	.149	.154	3.66	3.78	3.91	J	.100	.130	.160	2.54	3.30	4.06
F	.990	.994	.998	25.15	25.25	25.35	K	.850	.875	.900	21.59	22.23	22.86
H		.050			1.27								
M		.220			5.59		QUALITY CONFORMANCE INSPECTION, PART 2						
QUALITY CONFORMANCE INSPECTION, PART 1							G	.248	.249	.250	6.30	6.32	6.35
A			3.000			76.20	L			1.000			25.40

NOTES :- a METRIC EQUIVALENTS (TO THE NEAREST .01 MM) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 MM.
b ANY ORIENTATION OF PINCH-OFF IS PERMITTED.
c BASE MATERIAL SHALL BE IN ACCORDANCE WITH MIL-M-14, TYPE CFG.
d TOP CAP INSULATING WAFERS SHALL BE PER L-P-513.

FIGURE 1. Outline drawing of electron tube type 8467

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